5. Physical Hazards

Introduction

The hazards that the worker encounters in the work environment can be classified into three major groupings. The first grouping includes the toxic hazards that are a result of a chemical action on the human system. These toxic hazards were covered in the last chapter. The second major classification involves physical hazards. Physical hazards are a result of a physical force that is exerted on the human body by some action within the work environment. A third classification of hazards, which we shall call other hazards, includes psychological and biological stresses placed on the human system.

In this chapter, we will discuss briefly the various types of physical hazards and other hazards that the worker encounters. The effect of physical hazards on the worker may be twofold. Certain physical hazards may damage the health of the worker. Also, the worker may receive traumatic injury as a result of exposure to a physical hazard. Those physical hazards that produce the possibility of traumatic injury are generally considered to be within the realm of responsibility of the safety professional. However, the traumatic injury may be secondary to a toxic hazard or a physical hazard that exists in the work environment. For example, exposure to many of the aliphatic halogenated hydrocarbons can disorient a worker, thus making the worker subject to potential traumatic injury. This situation may occur even though the exposure level of the toxic substance is insufficient to cause a chronic or acute toxic reaction in the worker.

In the following paragraphs, the various types of physical and other hazards will be discussed. For each hazard, its characteristics and effect on the human system will be covered. Where protection mechanisms of the human system are brought into action, those protective mechanisms will also be presented. Each of the following hazards will be covered:

Physical Hazards

A. Noise
B. Vibration
C. Ionizing and nonionizing radiation
D. Thermal
E. Mechanical
F. Pressure
G. Illumination
H. Traumatic

Other Hazards

A. Biological
B. Psychological

Physical Hazards--Noise

When one first enters an industrial plant, the first impression received is usually that of the presence or absence of high levels of noise. Certain operations are noisy by nature.

The level of noise in an industrial operation can constitute a physical hazard to the exposed workers. The extent of hazard depends not only on the overall noise level, but also on the time period for which the worker is exposed, the frequency of the noise to which the worker is exposed, and the type of noise: i.e., whether it is continuous or intermittent. All these factors must be considered when determining if a noise hazard exists. A more thorough discussion of the determination of the existence of a noise hazard exposure will be presented in the section related to noise.

The existence of a noise exposure can have various effects on the worker. First, the worker who is exposed to a high-level noise for a short period of time can exhibit a temporary hearing threshold shift. This hearing threshold shift is a loss in hearing that can be recovered within a short time after removal of the noise source. In general, most of this recovery occurs within one to two hours of exposure, with complete recovery occurring in approximately 14 to 16 hours. If the worker is continually exposed to the same excessive noise for a long period of time, the temporary hearing threshold shift can become permanent. This permanent hearing loss cannot be recovered by the worker.

Another effect of excessive noise levels is its resultant interference with communication between workers. This interference with communication can be an annoyance to the worker and may result in a lowering of the efficiency of the operation. There also exists the possibility of traumatic injury because of this loss of communication.

Certain temporary physiological changes can occur in the human body when it is exposed to excessive noise levels. For example, the heart rate and blood pressure tend to increase. At present there are no data to indicate that these temporary physiological changes result in any chronic damage to the worker's system.

The temporary hearing loss exhibited when exposed to excessive noise is in itself a protective mechanism. It can serve as a warning to the worker who is temporarily exposed that unless something is done a more permanent hearing loss may be incurred. In addition, when more permanent hearing loss begins to occur, it can be diagnosed in the 3000-6000 Hz frequency range, with most affected persons showing a loss or "dip" at 4000 Hz. If high-level exposures are continued, the loss of hearing will further increase around 4000 Hz and spread to lower frequencies. Periodic audiometric tests can identify threshold shifts in the 3000-6000 Hz frequency range, thus providing for

corrective action prior to any hearing loss that might affect the worker's ability to understand voice communication.

Physical Hazards--Vibration

Vibration is often closely associated with noise. One of the reasons for this close association is the fact that generally if a vibration is present, a noise is also present. However, the noise that is present may not be at a level that can cause damage to the worker's hearing, while the vibration may be serious enough to merit concern.

Vibration results in the mechanical shaking of the body or parts of the body. These two types of vibration are called whole-body vibration and segmental vibration. Vibration originates from a mechanical motion generally occurring at some machine or series of machines. This mechanical vibration can be transmitted directly to the body or body part, or it may be transmitted through solid objects to a worker located at some distance away from the actual vibration.

The effect of vibration on the human body is not totally understood. Research has only begun to indicate where problems might exist. Initial research indicates that whole-body vibration increases the physiological activity of the heart and respiration. The results have also shown that there is an inhibition of tendon reflexes as a result of vibration. There seems to be reduced ability on the part of the worker to perform complex tasks, and indications of potential damage to other systems of the body also exist.

More research has been done in the area of segmental vibration, such as that received when using a pneumatic hammer. One recognized indication of the effect of segmental vibration is impaired circulation to the appendage that has been named Raynaud's Syndrome. Such segmental vibration can also result in the loss of the sense of touch in the affected area. There have also been indications of decalcification of the bones in the hand as a result of vibration transmitted to that part of the body. In addition, muscle atrophy has been identified as a result of segmental vibration.

As with noise, the human body can withstand short-term vibration even though this vibration might be extreme. In addition, the dangers of vibration are related to certain frequencies that are resonant with various parts of the body. Vibration outside these frequencies is not nearly so dangerous as vibration that results in resonance.

Physical Hazards--Ionizing and Nonionizing Radiation

Ionizing and nonionizing radiation are becoming increasingly prevalent in the industrial work environment. Ionizing radiation results from electromagnetic radiation with energy sufficient to cause the loss of an electron from the matter with which it interacts. Nonionizing radiation is caused by rays from the electromagnetic spectrum having energy which is insufficient to cause the loss of an electron. Ionizing radiation includes X-, gamma, alpha, beta, and neutron rays. Nonionizing radiation is caused by

ultraviolet, infrared, laser, and microwave rays from the electromagnetic spectrum.

The effects of nonionizing radiation on the human are not well documented. However, some major effects that have been identified include damage to the eye and its ability to function, damage to the reproductive system, and burns of the skin. Ionizing radiation can produce skin burns as well as deep tissue burns for certain types of ionizing radiation. Ionizing radiation can also produce chronic effects on the human system. For example, ionizing radiation has certain carcinogenic and gene-damaging properties that can affect the long-term health of the exposed individual and potentially his or her progeny. Ionizing radiation can occur as both an external and an internal exposure in the human. External exposure can occur as a result of unprotected proximity to gamma rays or X-rays. Internal exposure results from inspired radioactive material that can cause tissue damage in the lungs and transmit damage throughout the human system by way of the blood stream.

The body has very little protection against radiation. The skin acts as a protective mechanism for small doses of both ionizing and nonionizing radiation. The normal phagocytic action that occurs in the lungs for respirable dust is not nearly so effective a protective mechanism for inspired radioactive materials as it is for other respirable dusts.

Physical Hazards--Thermal

Workers may be exposed to thermal hazards involving extreme heat, extreme cold, or rapid changes in temperature. Most exposures involve workers in high temperature areas. However, those exposures related to cold and rapid change in temperature can also exist.

There are various effects of high temperature on the worker as well as a number of factors that affect the individual's response to exposure to high temperatures. Among the important factors are the age of the individual, the surface-area-to-weight ratio of the individual, and the acclimatization that the individual has attained. The ability to withstand high temperatures while performing work lowers in older workers. As an individual becomes more obese, his ability to withstand temperatures is also lowered. Individuals working in a hot area will be more effective when they have had a chance to become acclimatized to the heat. Another factor that affects the human's response to heat is the amount of physical activity performed while subjected to the hot environment. As the physical activity increases, the time that a given temperature can be tolerated without adverse effect decreases.

Certain physiological changes in the body result from exposure to extreme heat. The first of these is the loss of salt that occurs because of perspiration. In addition to this loss of salt, a loss of body water and dehydration occur. The sweat glands can exhibit fatigue after prolonged exposure to high temperature. Also, because of the dilation of blood vessels near the surface of the skin, a pooling of blood in the extremities can occur.

The physiological changes in the body can result in various heat illnesses. These illnesses are as follows:

A. Heat cramps--painful spasms of the muscles as a result of hard work during exposure to high temperature.

B. Heat exhaustion--the worker experiences fatigue, nausea, and headache. The worker may lose consciousness from heat exhaustion.

C. Heat stroke--generally results in mental confusion or loss of consciousness and can result in convulsions and coma if left untreated. Shock can result because blood pools in the blood vessels of the extremities.

D. Skin rash--a body rash resulting from the inflammatory reaction of the skin to perspiration and the plugging of the sweat glands.

Exposures to extreme cold can result in frostbite occurring in exposed areas of the body. In addition, long-term exposure to cold temperatures can lower the core temperature of the body, thus presenting a more acute danger to human life.

The human body can regulate its internal body temperature within narrow limits by perspiration and dilation and constriction of the blood vessels. The variation in blood vessel size occurs to cause blood to flow to the surface for cooling in the case of high temperature and to restrict flow in the case of low temperature. In addition, after exposure to high temperatures, the worker can become acclimatized, thus allowing the body to withstand hard work at high temperatures with less chance of adverse effects. The muscle movement or shivering that occurs in cold temperature is a reaction of the body to attempt to generate heat within the body tissue.

Physical Hazards--Mechanical

Mechanical hazards can occur when stress is placed on the musculoskeletal system beyond the limits that the system can endure. This stress can be the result of a blow, a constant pulling, or a pushing on the body structure. Mechanical stress can also be caused as a result of fatigue in the muscles that results in the worker performing tasks in an improper manner, thus causing undue stress to be placed on the body structure.

Mechanical hazards can result in traumatic injury. For example, the most likely effect of a blow to the body is a traumatic injury. Another example is the worker who is required to lift heavy objects repeatedly, thus becoming fatigued. As the worker becomes fatigued, the probability increases that he will drop one of the objects or cause damage to another part of his body by improper lifting.

Chronic disabilities can result from mechanical hazards. Workers who are required to maintain a particular position for prolonged periods of time can develop chronic ailments of the vertebrae or joints that are affected by the required position.

Fatigue is a protective mechanism the body employs to prevent mechanical damage from occurring. If the worker can heed the fatigue signals and rest, the likelihood of injury occurring is lessened. Body muscles, when allowed to rest, have great recuperative powers. To take advantage of these recuperative powers, however, it is necessary that the worker be given sufficient rest time for the muscles to regenerate.

Physical Hazards--Pressure

The exposure of workers to abnormal pressures is generally limited to a few occupational areas. Two of the most obvious of these are the individuals who work as sand hogs constructing tunnels under bodies of water, and individuals working in diving apparatus. In both of the above cases, the workers are exposed to higher-than-atmospheric pressures. It is possible that workers can be exposed to lower-than-atmospheric pressure. This may become more prevalent in the future as space travel expands.

At high pressures, workers may experience an acute syndrome called "oxygen poisoning." Nitrogen poisoning, which has a narcotic effect on the body, can also occur at high pressures. The most commonly known effect of high pressure is the necessity to decompress the workers from the high pressure to normal atmospheric pressure. If decompression is not handled properly, the worker develops decompression sickness, otherwise known as "the bends." High pressures can act on the fluids and tissue of the body to compress these fluids and tissue into the cavities of the body, such as the sinuses.

Low pressure can result in an oxygen deficiency to the body; also, the expansion of gases into the cavities of the body is noticed under low pressure. Such expansion of gases can cause severe pain. In addition, the worker entering a low-pressure area is, in effect, being decompressed and is thus subjected to the same decompression sickness that occurs during decompression from high pressures.

The body attempts to adjust for low pressures by increasing the number of red blood cells in the blood at higher altitudes, thus allowing for increased oxygen transported to the tissues. At low pressure, respiration will increase, thus bringing more oxygen to the respiratory system. In addition, individuals can become acclimatized to low pressures, especially in the case of low pressure at high altitudes.

Physical Hazards--Illumination

Inadequate illumination is a hazard to the worker and can cause a number of detrimental effects. Eyestrain can develop if the worker is required to perform close and exacting work with inadequate illumination. In addition to eyestrain, inadequate illumination can also cause errors and reduced

efficiency. Attempts to compensate for these errors and reduced efficiency can increase the chance for traumatic injuries. In poorly illuminated areas, the potential for accidents is high.

The body's basic protective mechanism to adjust for low illumination is dilation of the pupils. This mechanism does not react quickly; an individual subjected to a glare or a quick change in the amount of illumination will not react immediately. These changes in illumination and the presence of glare, therefore, result in the individual being subjected to a period of time during which vision is inadequate. During this period, the probability of traumatic injury increases greatly.

Physical Hazards--Traumatic

The number of potential causes of traumatic injury to the worker are legion. Some of these, such as changes in illumination and fatigue, have been mentioned. Four general causes of traumatic injury can be stated. In some cases, these causes may be secondary to a primary cause. The first of these is the potential for fire and explosion. The primary result of a fire or an explosion is injury to the individuals in close proximity. The second potential cause of traumatic injury is through mechanical means. This can occur when an individual interfaces with a piece of machinery such as a radial saw or punch press. A third major cause of physical injury involves the interface of the individual with electrical systems. Finally, the worker may be injured as a result of a structural failure, such as when a ladder breaks or the worker falls on a slippery floor.

The obvious effects of traumatic injury are the physical damage that the body incurs as well as the possibility of death. Organizationally, the effects of traumatic injury are felt in terms of property damage and the cost of lost-time injuries on the part of the workers.

As opposed to the health hazards that are faced, the individual worker has adequate protection to avoid many of the potential traumatic injuries that are encountered. This protection exists in the proper use of the senses of the body. The worker must learn to identify through sight, sound, smell, touch, and taste those factors in the work environment that can result in an unsafe situation. However, the amount of concentration required on the part of the worker to do this consistently makes these protective mechanisms only partially useful.

Other Hazards--Biological

Biological hazards consist of exposure to bacteria, viruses, and parasites. This exposure can be a direct result of the work being performed or the result of unhealthy conditions in the work environment. The obvious example of exposure as a result of the work being performed is the worker in the hospital whose job requires contact with various communicable diseases. Exposure in the work environment can result from unsanitary conditions in rest rooms, eating areas, and locker rooms. The obvious result of biological exposures is the illness of the worker and the transmission of the disease to other associated workers.

The body has an internal mechanism that creates white blood cells to fight infections. In addition, the individual can be protected from many diseases by immunization, which produces antibodies that protect against possible future infection of a particular disease. Since immunization is disease specific, it is not a cure-all protective device.

Other Hazards--Psychological

Consideration has only recently been given to psychological job stress. Workers may be affected psychologically by pressures that are inherent within the job or pressures that exist outside the job. Endogenous pressures include such things as unrealistic production schedules, demanding management, and unclear directions concerning the work to be performed. Exogenous factors include the worker's peer group relationship, the general atmosphere in which the work is accomplished, and the person's satisfaction with his home life.

Research is only beginning in this area of job stress. No conclusions have been reached, although many hypotheses have been put forth. There currently is a feeling that psychological stress can result in physiological changes in the body. Behavior changes are also hypothesized. It is felt that some workers develop somatic complaints and become ill as a result of psychological stresses. There does appear to be some correlation between individuals under psychological stress and the accident rate encountered by individuals in similar jobs.

Man adjusts and makes decisions. As a result of this fact, individuals will tend to modify their behavior patterns when faced with psychological stress. This modification of behavior patterns can result in a lower stress on the worker involved. In addition, workers faced with undue psychological stress will tend to self-select out of the job, thus removing themselves from the hazard. This, however, does not prohibit other workers from entering into the same job and facing the same psychological stress.

Summary

It is evident from the previous discussion that a given worker can face many potential physical hazards while performing his work. In any given situation, not all of these hazards are important. In fact, certain groups of jobs encounter only a few of the hazards that have been mentioned. It is the job of the industrial hygienist or safety professional to identify the hazards that the worker faces on a given job and to take action to control the worker's exposure to these hazards.

Although certain protective mechanisms do help the human system adapt to its surroundings, these protective systems operate only within a small range. It should be remembered that a hazard exists when the stress placed on the human system cannot be compensated for by the internal mechanisms of the system without resulting in strain. In such a situation, it is the responsibility of the industrial hygienist or safety professional to remove the stress or to cause it to be lowered to such a level that the compensating mechanism within the human system will provide protection without strain on the workers.